

Python Code

```
1  import random
2
3  def guessing_game():
4      # 1. Generate a random number between 1 and 100
5      secret_number = random.randint(1, 100)
6      attempts = 0
7
8      print("Welcome to the Number Guessing Game!")
9      print("I am thinking of a number between 1 and 100.")
10
11     # 2. Start a loop to let the user keep guessing
12     while True:
13         try:
14             # 3. Get input from the user and convert it to an integer
15             guess = int(input("Take a guess: "))
16             attempts += 1
17
18             # 4. Check if the guess is too low, too high, or correct
19             if guess < secret_number:
20                 print("Too low! Try again.")
21             elif guess > secret_number:
22                 print("Too high! Try again.")
23             else:
24                 print(f"🎉 Congratulations! You guessed the number in {attempts} attempts!")
25                 break # Exit the loop when they win
26
27         except ValueError:
28             # Handle the error if the user types letters instead of a number
29             print("Invalid input. Please enter a valid whole number.")
30
31     # Run the game
32     guessing_game()
33
```